

**Karnataka
Physics- 2017**

General Instructions:

- 1) All the parts are compulsory.**
- 2) Answer without relevant diagram /figure/circuit wherever necessary will not carry any marks.**
- 3) Direct answers to numerical problems without detailed solutions will not carry any marks.**

I. PART-A

Answer the following questions:

1. State Coulomb's Law.

Sol. 'The Electrostatic force of attraction or repulsion between two stationary point charges. Separated in free space or air is directly proportional to the product of magnitude of charge. On them and inversely proportional to the square of the distance between them and the

Force acts along the line joining the two point charges.'

2. Define mobility of electron.

Sol. It is the drift velocity of free electrons per unit electric field.

3. What is the significance of Lenz's Law?

Sol. It gives the direction of the induced current.

4. Define displacement current.

Sol. It is the current due to charging electric flux with time is known as displacement current.

5. Write one application of microwave.

Sol. Microwave are used in

- (1) Radar system for aircraft navigation
- (2) For heating in microwave oven
- (3) Measure the speed of automobile, cricket ball and tennis ball.

6. How the power of a lens is related to its focal length?

Sol. Inversely Or $P = \frac{1}{f}$

7. Write the expression for de-Broglie wavelength of a particle.

Sol. The de-Broglie's equation is $\lambda = \frac{h}{mv}$,

Where λ is the de-Broglie wavelength, m the mass and v the velocity of the particle.

8. What is the conclusion of Davisson-Germer experiment on the nature of electron?

Sol. Wave nature of electron.

9. Write the SI unit of activity.

Sol. Decay/sec.

10. What is transducer in communication system?

Sol. A device that converts one form of energy (signal) into another.

II. PART-B

11. Write any two properties of electric field lines.

Sol. (1) They never intersect each other.

(2) Electric field lines start from positive charge and ends at negative charge.

12. On what factors does the capacitance of a parallel plate capacitor depends?

Sol. Capacitance of a parallel plate capacitor depends directly on the area of cross section of the plate, dielectric constant of medium between the two plates and inversely proportional to the separation between the plates.

13. State and explain Ohm's Law.

Sol. Ohm's Law: When all the physical conditions surrounding the conductor remain Unaltered, the electric current in a conductor is directly proportional to the potential Difference between the ends of the conductor.

$$I \propto V$$

$$I = KV$$

Here, K is an electrical conductance.

$$V = IR$$

Here, $R = \frac{1}{K}$ is an electrical resistance of the conductance.

14. Define the terms:

(1)Declination

(2)Inclination or Dip.

Sol. (1) The angle between the geographic meridian and magnetic meridian.

(2) The angle between earth's total magnetic field and its horizontal component.

15. State and explain Faraday's Law of electromagnetic induction.

Sol. The magnitude of the induced emf in a circuit is directly proportional to the rate of change of magnetic flux linked with the circuit.

If $d\phi_B$ is the change in magnetic flux in time dt , then induced emf is

$$\varepsilon = \frac{d\phi_B}{dt}$$

16. Name the type of lens which is used to correct.

(1)Myopia

(2)Hypermetropia.

Sol. (1) Concave lens

(2) Convex lens

17. What is NAND gate? Give its logic symbol.

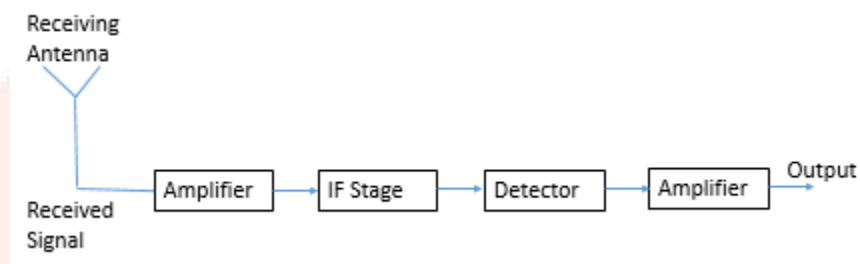
Sol. It is an AND gate which is followed by a NOT gate.

Inputs A	B	Output $Y = \overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0



18. Draw the block diagram of a AM receiver.

Sol.



III. PART-C

19. Derive the relation between electric field and electric potential due a point charge.

Sol. Consider a point Charge '+d'. Let 'P' be a point initially. Let 'A' be a point inside the field region let '+1C' be a unit positive charge moved from the point at infinity to point 'A'. Let 'A' be at a distance of 'r' from the given point charge. Let 'B' be another point at distance of 'dr' from 'a' towards the point charge +q.

The work done in moving positive test Charge from A to B, against the force of repulsion is $dW = -Fdr$. The -ve sign indicates that the work is done against the direction of the force and dr is the displacement of +1 C of test charge in the direction opposite to the electric field.

But electric potential $dV = \frac{dW}{+q_0}$ and electric field $E = \frac{F}{+q_0}$

When $q_0 = +1 C$ then $dW = dV, E = F$

Hence $dV = -Edr$ or $E = -\frac{dV}{dr}$

i.e., Electric field intensity at a point is the negative potential gradient at the point and electric field intensity is in the direction of decreasing electronic potential.

20. Derive the expression for energy stored in a charged capacitor.

Sol. Let 'C' be the capacitance of a parallel plate capacitor. By definition $C = \frac{Q}{V}$.

By definition amount of work done to raise the charge by 'dq' is $dQ = Vdq$

$$\text{i.e. } dW = \frac{q}{C} dq$$

Where 'q' is the instantaneous value of charge on the conductor. Total work done in charging the capacitor is given by,

$$\Rightarrow W = \int_{q=0}^{q=Q} dW$$

$$\Rightarrow W = \int_{q=0}^{q=Q} \left(\frac{q}{C} \right) dq$$

$$\Rightarrow W = \frac{1}{C} \left[\frac{q^2}{2} \right]_0^Q$$

Or $W = \frac{1}{2} \frac{Q^2}{C}$, amount of work done is stored in the form of energy

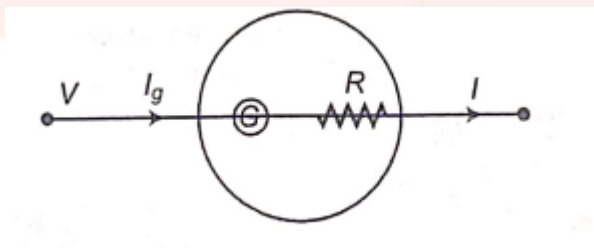
21. Explain with circuit diagram how to convert galvanometer into a voltmeter.

Sol:

To convert a galvanometer into a voltmeter, a high resistance is connected in series with the coil. The galvanometer with this modification is called a voltmeter. The value of the high resistance R to be connected in series with the coil depends on the maximum potential difference to be measured. If I_g be the current for a full scale deflection then,

$$V = I_g (G + R)$$

G is the resistance of the galvanometer.



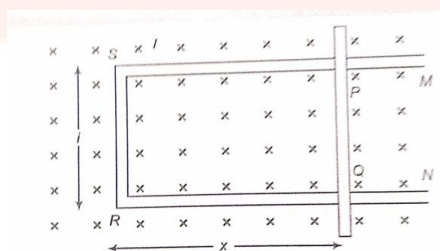
$$\therefore R = \frac{V}{I_g} - G$$

$$I_g \propto V$$

Since, G and R are constants, the scale can be graduated to read potential differences directly. A voltmeter is used for measurement of potential difference. It should be connected in parallel in a circuit.

22. Derive the expression for motional emf induced in a conductor moving in a uniform magnetic field.

Sol:



Let 'PQ' represent a conductor moving with a speed 'v' at right angles to the magnetic flux density 'B' pointing perpendicular to the sheet of paper and pointing towards the paper. Let RQ = x be the distance covered by the conductor. Magnetic flux $\phi_B = Blx$

Induced emf

$$e = -\frac{d(\phi_B)}{dt} = -\frac{d}{dt}(Blx)$$

$$e = -Bl \frac{dx}{dt}$$

but,

$$\frac{dx}{dt} = v$$

$$\therefore e = Blv$$

Note: The Lorentz force acts on a charge in a conductor and acts along PQ. Work done in moving a charge 'q' from P to Q is

$$W = (\text{Lorentz force}) \times (\text{displacement of the charge 'q'})$$

$$W = (Bqv)l$$

$$\text{By definition electric p.d} = \frac{W}{q} = \frac{Bqvl}{q} = Blv$$

Equation (i) and (ii) are identical.

23. What is transformer? Mention two sources of energy loss in a transformer.

Sol: It is a device used to increase or decrease the AC voltage.

Sources of Energy Loss:

- (i) Magnetic flux leakage loss.
- (ii) Ohmic loss/loss due to the resistance of the windings (turnings)/Joule's heat loss.
- (iii) Eddy current loss.
- (iv) Hysteresis loss.

24. Mention any three applications of Polaroid's.

- Sol:**
- (i) In sunglasses
 - (ii) In liquid crystal display
 - (iii) To view 3-D pictures
 - (iv) Automobile head lights

25. Write any three experimental observations of photo electric effect.

- Sol:**
- (i) The photoelectric emission is an instantaneous process.
 - (ii) For every photo emissive material, there exists a minimum frequency of radiation below which there is no photoelectric emission. This minimum frequency is called threshold frequency (ν_0).
 - (iii) Above the threshold frequency, the photocurrent is directly proportional to the intensity of incident radiation.
 - (iv) Above the threshold frequency, the maximum kinetic energy of the photo electrons is directly proportional to the frequency of incident radiation.
 - (v) The photoelectric current increases with increase in collector plate voltage and reaches a saturation value.
 - (vi) The photocurrent decreases with increase in negative potential and reaches zero at a negative potential known as stopping potential.

26. Give any three differences between n-types and p-types semiconductors.

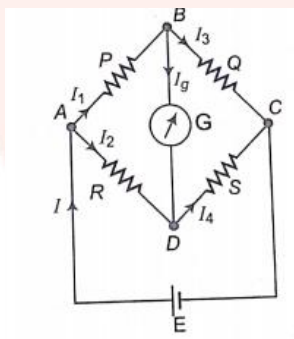
Sol:

S.No.	n-type semiconductor	p-type semiconductor
(i)	Obtained by doping pentavalent impurity to a pure semiconductor.	Obtained by doping trivalent impurity to a pure semiconductor.
(ii)	Majority charge carriers are free electrons.	Majority charge carriers are holes.
(iii)	Minority charge carriers are holes.	Minority charge carriers are free electrons.
(iv)	At 0 K only free electrons are available for conduction.	At 0 K only holes are available for conduction.
(v)	Donor energy level is slightly below the bottom of conduction band.	Acceptor energy level is slightly above the top of valance band.
(vi)	Impurity atom is called donor impurity.	Impurity atom is called acceptor impurity.

IV. PART-D

27. Deduce the condition for balance of a Wheatstone's bridge using Kirchhoff's rules.

Sol:



Let I_1, I_2, I_3, I_4 and I_g be the currents through the resistances, P, R, Q, S and G respectively as shown in figure. Apply KVL to the mesh ABDA, we get

$$I_1 P + I_g G - I_2 R = 0 \text{ -----(i)}$$

Apply KVL to the mesh BCDB, we get

$$I_3 Q - I_4 S - I_g G = 0 \text{ -----(ii)}$$

The network is balanced, if $I_g = 0$ then

At B, $I_1 = I_3$

At D, $I_2 = I_4$ -----(iii)

Equation (i) and (ii) becomes,

$$I_3 P = I_4 R \text{ -----(iv)}$$

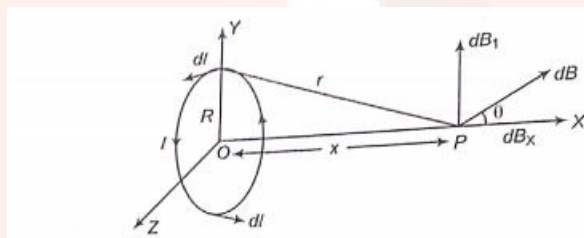
$$I_3 Q = I_4 S \text{ -----(v)}$$

Equation (iv) and (v), we get

$$\frac{P}{Q} = \frac{R}{S}$$

28. Derive the expression for magnetic field at a point on the axis of a circular current loop.

Sol:



$I \rightarrow$ Current in the loop,

$R \rightarrow$ Radius of the loop,

$X\text{-axis} \rightarrow$ Axis of the loop,

$x \rightarrow$ Distance OP

$dl \rightarrow$ Conducting element of the loop

According to Biot-Savart's law, the magnetic field at P is

$$db = \frac{\mu_0 I |dl \times r|}{4\pi r^3}$$

$$r^2 = x^2 + R^2$$

$$|dl \times r| = r dl$$

$$\therefore dB = \frac{\mu_0}{4\pi(x^2 + R^2)}$$

i.e, dB has two component- $-dB_x$ and dB_1 , dB_1 is cancelled out and only the x-component remains.

$$\therefore dB_x = dB \cos \theta$$

$$\cos \theta = \frac{R}{(x^2 + R^2)^{\frac{1}{2}}}$$

$$\therefore dB_x = \frac{\mu_0 Idl}{4\pi} \frac{d}{(x^2 + R^2)^{\frac{3}{2}}}$$

Summation of dl over the loop is given by $2\pi R$

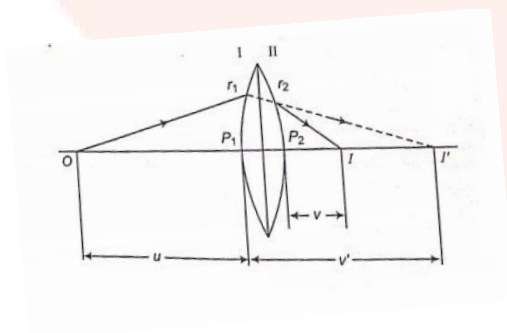
29. Write any five properties of ferromagnetic materials.

Sol:

- (i) They are strongly attracted by magnetic field.
- (ii) They are strongly magnetized in the direction of the external magnetic field.
- (iii) The magnetic field lines are highly concentrated inside the substance.
- (iv) χ is large positive.
- (v) $\mu_r \gg 1$
- (vi) Above the curie temperature, its susceptibility varies inversely proportional to the excess of temperature
- (vii) They exhibit magnetic hysteresis.

30. Derive Lens Maker's formula for a convex lens.

Sol: Lens maker's formula



Let r_1 and r_2 be the radii of curvature of the thin lens.

Let O be the point object at a distance u from the pole of the first curved surface of the lens. Real image is formed at I' , at a distance v' from the pole P_1 . This image is formed in the denser medium. We know that from the refraction the refraction formula,

$$\frac{n_1}{-u} + \frac{n_2}{v} = \frac{n_2 - n_1}{r}$$

$$\frac{n_1}{-u} + \frac{n_2}{v'} = \frac{n_2 - n_1}{r_1} \text{----- (i)}$$

Where n_1 is the refractive index of rarer medium and n_2 that of the denser medium ($n_2 > n_1$). For the second surface, real image at I' will serve as a virtual object and the distance is taken as -ve. The object space is the lens medium for refraction through the second curved surface. Final image is formed in air at I and at a distance of ' v ' from P_2

$$\frac{n_1}{-u} + \frac{n_2}{v} = \frac{n_2 - n_1}{r}$$

$$\frac{n_1}{v'} + \frac{n_2}{v} = \frac{n_2 - n_1}{-r_2} \text{----- (ii)}$$

Adding (i) and (ii), we get

$$\frac{n_1}{-u} + \frac{n_1}{v} = (n_2 - n_1) \left(\frac{1}{r_1} + \frac{1}{r_2} \right) \text{----- (iii)}$$

$$\frac{1}{-u} + \frac{1}{v} = \frac{n_2 - n_1}{n_1} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

When $u = \infty, v = f$

When $u = f, v = \infty$

$\therefore$ The term on the L.H.S can be replaced by $\frac{1}{f}$ where f is the focal length of lens.

$$\frac{1}{f} = \left(\frac{n_2 - n_1}{n_1} \right) \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \text{----- (iv)}$$

The equation (iv) is called the lens maker's formula.

$$\frac{1}{f} = \frac{n_2 - n_1}{n_1} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

Where,

$$1^n 2 = \frac{n_2}{n_1}$$

(2) For radius of curvature the letter R may be used.

31. State radioactive decay Law. Derive $N = N_0 e^{-\lambda t}$ for a radioactive element.

Sol:

The number of atoms disintegrated per second at any instant is directly proportional to the number of radioactive atoms actually present in the sample at that instant.

Mathematically, $\frac{dN}{dt} = -\lambda N$

Here N is the number of radioactive nuclei present in the sample at an instant.

$$\frac{dN}{dt} = -\lambda N$$

Where λ is disintegration constant.

$$\frac{dN}{dt} = -\lambda dt$$

$$\frac{dN}{N} = -\lambda \int dt$$

$$\log_e N = -\lambda t + C$$

$$C = \log_e N_0$$

$$\text{At } t = 0, N = N_0$$

$$\text{Thus, } \log_e N = -\lambda t + \log_e N_0$$

$$\log_e \left(\frac{N}{N_0} \right) = -\lambda t$$

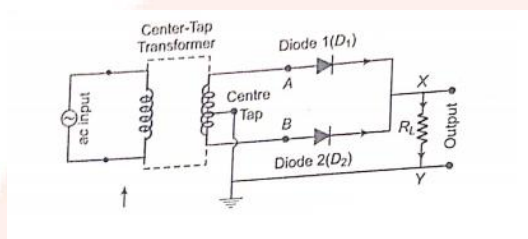
$$N = N_0 e^{-\lambda t}$$

32. What is rectification? With relevant circuit diagram and waveforms explain the working of $p-n$ Junction diode as a full wave rectifier.

Sol:

Full wave rectification: The process in which both the halves of a.c. are rectified to obtain a pulsating direct current is known as full wave rectification.

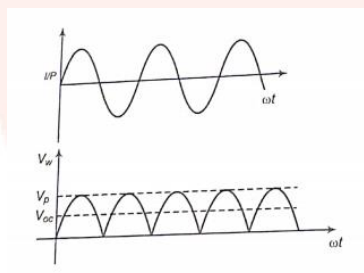
A transformer with a centre tap is used for a full wave rectification. Input contains ac. A minimum of two diodes are used at the secondary of the transformer. A load resistor is connected between the junction of the N type semiconductor and the centre tap.



During the first half cycle, diode D_1 will be forward biased and D_2 will be reverse biased so that the current flows from B to centre tap.

During the second half cycle, diode, D_2 will be forward biased while D_1 will be reverse biased. The current once again takes the same direction from B to centre tap. The I/P and the O/P

waveforms are shown in fig. The average DC is $2 \left(\frac{V_p}{\pi} \right)$



33. Two point charges $q_A = 3\mu C$ and $q_B = -3\mu C$ are located 20 cm apart in vacuum.

(a) What is the electric field at the mid-point O of the line AB joining the two charges?

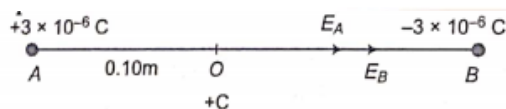
(b) If a negative test charge of magnitude $1.5 \times 10^{-9} C$ is placed at this point. What is the Force experienced by the test charge?

Sol:

Given,

$$q_A = 3 \times 10^{-6} \text{ C}, q_B = -3 \times 10^{-6} \text{ C}, r = 0.20 \text{ m}$$

(a) Electric field at the mid-point 'O'



$$E = \frac{1}{4\pi \epsilon_0} \cdot \frac{Q}{r^2}$$

$$E = \frac{9 \times 10^9 \times Q}{r^2} \text{ Vm}^{-1}$$

Here,

$$E_A = E_B = \frac{9 \times 10^9 \times 3 \times 10^{-6}}{(0.10)^2}$$

$$E_A = E_B = \frac{27 \times 10^3}{10^{-2}} = 27 \times 10^5 \text{ Vm}^{-1}$$

The direction along AB or OB.

$$\therefore \text{Net electric field at } = E_A + E_B$$

$$5.4 \times 10^6 \text{ Vm}^{-1} \text{ along OB}$$

(b) Net force on 1.5×10^{-9} placed at 'O'

$$F = q_0 E_0$$

$$= -1.5 \times 10^{-9} \times 5.4 \times 10^6 \text{ N}$$

$$= 8.1 \times 10^{-3} \text{ N in the direction along OA}$$

34. When two resistors are connected in series with a cell of emf 2V and negligible internal resistance, a current of 2/5 A flows in the circuit. When the resistors are connected in parallel the main current is 5/3 A. Calculate the resistance.

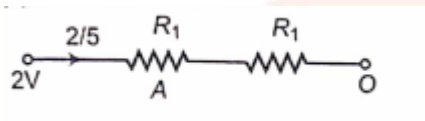
Sol:

Given,

$$E = 2V, r = 0, I_s = \frac{2}{5} A$$

$$I_p = \frac{5}{3} A \quad R_1 = ? \quad R_2 = ?$$

(a) Resistors in series:

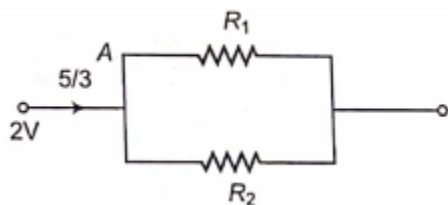


$$I_s = \frac{E}{R_s + r}$$

$$\frac{2}{5} = \frac{2}{(R_1 + R_2) + 0}$$

$$\Rightarrow R_1 + R_2 = 5$$

(b) Resistors in Parallel:



$$I_p = \frac{E}{R_p + r}$$

$$\frac{5}{3} = \frac{2}{\left(\frac{R_1 R_2}{R_1 + R_2} + 0 \right)}$$

$$\Rightarrow \frac{R_1 R_2}{R_1 + R_2} = \frac{6}{5}$$

$$\therefore R_1 + R_2 = 5, R_1 R_2 = 6$$

$$\text{And } R_1 + R_2 = \sqrt{(5)^2 + 4(6)} = 1$$

Solving

$$R_1 + R_2 = 5$$

$$R_1 - R_2 = 1$$

$$2R_1 = 6$$

$$R_1 = 3\Omega \text{ and } 3 - R_1 = 1$$

$$R_2 = 2W$$

Hence $R_1 = 3\Omega, R_2 = 2W$

35. A source of alternating emf of $220V - 50Hz$ is connected in series with a resistance of 200Ω an inductance of $100mH$ and a capacitance of $30\mu F$. Does the current lead or lag the voltage and by what angle?

Sol: Given,

$$V_{rms} = 200V, f = 50Hz, R = 200W$$

$$L = 100 \times 10^{-3} = 0.1H$$

$$C = 30 \times 10^{-6} F = 3 \times 10^{-5} F$$

$$\text{Inductive reactance, } X_L = 2\pi fL$$

$$X_L = 2 \times 3.142 \times 50 \times 0.1W$$

$$X_L = 31.42W$$

$$\text{Capacitive reactance, } X_C = \frac{1}{2\pi fC}$$

$$X_C = \frac{1}{2 \times 3.142 \times 50 \times 3 \times 10^{-5}}$$

$$X_C = \frac{1000}{9.426} = 106.08\Omega$$

$$\text{Reactance of the circuit } X_L - X_C = 31.42 - 106.08$$

$$X_L - X_C = -74.66W$$

$$\Rightarrow \tan \phi = \frac{X_L - X_C}{R}$$

$$\tan \phi = \frac{-74.66}{200} = 0.3733$$

$$\text{Hence } \phi = -20^\circ 30'$$

The current leads the voltage by $20^\circ 30'$

36. Light of wavelength $6000 \text{ \AA}$ is used to obtain interference fringes of width 6mm in a Young's double slit experiment. Calculate the wavelength of light required to obtain fringe of width 4mm when the distance between the screen and slits is reduced to half of its initial value.

Sol:

Given,

$$\lambda = 6800 \times 10^{-10} \text{ m} = 6 \times 10^{-7} \text{ m}$$

$$\beta = 6 \times 10^{-3} \text{ m}, \beta' = 4 \times 10^{-3} \text{ m}$$

$$\lambda' = ?$$

$$D' = \frac{D}{2}$$

$\Rightarrow$ fringe width,

$$\beta = \frac{\lambda D}{d}$$

Hence,

$$\frac{\beta'}{\beta} = \frac{\lambda' D'}{d} \times \frac{d}{\lambda D}$$

$$\frac{\beta'}{\beta} = \frac{\lambda' D'}{d} \times \frac{d}{\lambda D}$$

$$\beta' = \frac{\lambda' \beta}{2\lambda}$$

$$\lambda' = \frac{2\lambda\beta'}{\beta}$$

$$\lambda' = \frac{2 \times 6 \times 10^{-7} \times 4 \times 10^{-3}}{6 \times 10^{-3}}$$

$$\lambda = 8 \times 10^{-7} \text{ m}$$

$$\lambda' = 8000 \text{ \AA}$$

The wavelength required to produce fringes of width 4 mm is $8000 \text{ \AA}$

37. The first member of the Balmer series of hydrogen atom has wavelength of $6563 \text{ \AA}$. Calculate the wavelength and frequency of the second member of the same series. Given $c = 3 \times 10^8 \text{ ms}^{-1}$.

Sol: Given, $\lambda_0 = 6563 \text{ \AA} = 6.563 \times 10^{-7} \text{ m}$

$$\lambda_\beta = ? \quad f_\beta = ?$$

$$c = 3 \times 10^8 \text{ ms}^{-1}$$

$$\Rightarrow \frac{1}{\lambda_1} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

For 1 member of Balmer series, $n_1 = 2, n_2 = 3$

$$\therefore \frac{1}{6.563 \times 10^{-7}} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right)$$

$$\frac{10^7}{6.563} = R \left(\frac{5}{36} \right)$$

Hence,

$$R = \frac{36}{5} \times \frac{10^7}{6.563}$$

$$R = 1.097 \times 10^7 \text{ m}^{-1}$$

For II member of Balmer series

$$x_1 = 2, x_2 = 4$$

$$\therefore \frac{1}{\lambda_2} = 1.097 \times 10^7 \left(\frac{1}{2^2} - \frac{1}{4^2} \right)$$

$$\frac{1}{\lambda_2} = \frac{1.97 \times 10^7 \times 12}{64}$$

$$\lambda_2 = \frac{64}{1.097 \times 12 \times 10^7} \text{ m}$$

Hence,

$$\lambda_2 = 4.8617 \times 10^{-7} \text{ m}$$

or,

$$\lambda_2 = 4861.7 \text{ \AA}$$

And frequency,

$$Y_2 = \frac{c}{\lambda_2} = \frac{3 \times 10^8}{4.8617 \times 10^7}$$

$$Y_2 = 0.6171 \times 10^{15} \text{ Hz} = 6.171 \times 10^{14} \text{ Hz}$$

